

Human vs. LLM Performance in Data Extraction

Isaac Abraham
George R. Brown School of Engineering and Computing
Rice University
Houston, United States of America
Isaac.Abraham@rice.edu
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Abstract—This article compares the accuracy of data extraction from graphs using large language models, like Google’s Gemini Pro, xAI’s Grok, Anthropic’s Claude, OpenAI’s ChatGPT, and DeepSeek, to manual extraction with WebPlotDigitizer. The different methods of data extraction used by LLMs – fine-tuning and built-in reasoning architecture – are examined in this article. The study explores differences in precision using a complex visualization of a solid red line, revealing that LLMs are not adept at accurately extracting information from data graphs.

I. INTRODUCTION

Large language models (LLMs) are deep learning models trained to capture patterns, nuances, and deeper context in data. LLMs are trained on large amounts of data, enabling them to perform a variety of tasks with high accuracy, including summarizing documents, identifying patterns in essays, answering questions, and more [1]. Understanding data visualizations, such as bar and line charts, is essential for deriving insights from different variables, and this study will explore LLMs’ ability to do so. There are many methods that LLMs use to understand and extract data from visualizations. The most common are fine-tuning and built-in reasoning architectures.

II. HOW DO LARGE LANGUAGE MODELS EXTRACT DATA FROM GRAPHS

A. Fine-tuning

Fine-tuning is used when training a pre-trained model on a specific dataset to be applied to other data visualizations for analysis. The process of fine-tuning an LLM begins by gathering training examples. These examples are paired with the correct data extracted from them so that the large language model can learn the patterns [2]. In addition to these examples, the model should be guided to understand which data are extracted from each example. Afterward, the model undergoes a two-stage process of understanding the charts. The

first stage maps the visual features of the visualizations to the LLM’s word embedding space, which is how LLMs capture the semantic meaning of items [3]. The second stage involves teaching the model to follow instructions about extracting data from those data visualizations. The model is then iteratively trained, using the above processes. Finally, the model is tested and adjusted based on its performance.

B. Built-in Reasoning Architecture

Certain LLMs use the reasoning structures built into their design to break down complex graphs. This is typically what larger LLMs that are not specifically designed to extract data from visualizations use. The reasoning process begins with the LLM listing all numerical data that can be identified on the graph, on both axes. This data is then structured into a table with appropriate column headers and units [4]. The data is then sorted in descending order by numerical values to capture the trends in the information. The LLM then double-checks the values in the table to verify they were accurately extracted from the original visualization. Finally, the LLM uses comparisons (less than and greater than) to understand trends in the data.

III. LLM VERSUS MANUAL EXTRACTION

The most common data extraction algorithm used by LLMs is their native reasoning architecture. This is typically how newer LLM models, such as Claude 4.5 Sonnet, Gemini 2.5 Pro, and OpenAI, understand data from graphs. To assess the accuracy of this extraction technique, the built-in reasoning structure of Gemini 2.5 Pro, Grok, Claude Sonnet 4.5, ChatGPT 4.0, and DeepSeek will be compared with manually extracted data using WebPlotDigitizer. The data will describe the

relationship between glucose concentration (mg/dL) and time (minutes) for patients with diabetes A.

Fig. 1. Graph of the solid red line that will be extracted.

A. Google’s Gemini 2.5 Pro

Gemini originated from the merging of Google’s DeepMind and Brain research teams. The mission of Gemini is to be a universal, highly adaptable assistant. The built-in reasoning architecture Gemini uses is its native multimodal Visual Language Models (VLMs). Gemini’s VLMs combine computer vision and natural language processing capabilities to understand context, interpret content, extract specific data, identify trends, and generate code for charts [5].

The data extracted from Gemini 2.5 Pro’s built-in reasoning architecture and manually from WebPlotDigitizer, along with the percentage error, is listed in Table 1.

TABLE I. GEMINI SOLID RED LINE DATA EXTRACTION

t (min)	Data Extracted from Solid Red Line		
	Gemini (mg/dl)	WebPlotDigitizer (mg/dl)	Percent Error
0	70	56	20.00
25	65	64	1.56
50	110	107	2.80
75	120	112	7.14
100	105	107	1.87
125	115	120	4.17
150	95	107	11.21
175	90	108	16.67
200	85	100	15.00
225	80	78	2.56
250	75	74	1.35
275	75	83	9.64
300	65	73	10.96
325	70	54	29.63
350	70	66	6.06

B. xAI Grok

Grok is another large language model capable of extracting information from visuals. Grok was created by xAI with the mission of understanding the true nature of the universe. Grok is well equipped to handle textual information, with its

primary application on the social media platform X. It is also used to extract information from visuals. Grok’s multimodal model, with vision and language capabilities, is designed to extract structured data from images [6]. This enables Grok to extract data and generate analytical insights in tabular formats. Table 2 presents the data extracted from Grok, along with the manually extracted data and the percentage error.

TABLE II. GROK SOLID RED LINE DATA EXTRACTION

t (min)	Data Extracted from Solid Red Line		
	Grok (mg/dl)	WebPlotDigitizer (mg/dl)	Percent Error
0	95	56	41.05
10	105	57	84.21
20	160	61	162.29
30	240	67	258.20
40	295	86	243.02
50	315	107	194.39
60	310	107	189.71
70	275	109	152.29
80	230	114	101.75
90	190	117	62.39
100	165	107	54.20
110	155	98	58.16
120	150	118	27.12
140	140	109	28.44
160	145	107	35.51
180	135	112	20.53
200	125	100	25.00
220	130	81	60.49
240	115	81	41.97
260	195	71	174.64
280	110	82	34.14
300	120	73	64.38
320	115	60	91.67
340	105	61	72.13
350	95	66	43.94

C. Anthropic’s Claude Sonnet 4.5

Claude is a large language model created by the AI safety and research company Anthropic. Founded by former OpenAI executives, Anthropic was built to focus on the responsible development and

deployment of AI. The guiding principle of Claude is its constitutional AI. This feature ensures the large language model evaluates and refines its own outputs to produce the most accurate and ethical content [7].

Claude performs well at analyzing data from visualizations and conducting complex analyses on structured datasets. Claude does this by generating and running Python and JavaScript code to recreate the charts and summarize trends found in the visual data [8].

The data extracted from Claude’s code-generation analysis and from WebPlotDigitizer, along with the percentage error, are listed in Table 3.

TABLE III. CLAUDE SOLID RED LINE DATA EXTRACTION

t (min)	Data Extracted from Solid Red Line		
	Grok (mg/dl)	WebPlotDigitizer (mg/dl)	Percent Error
0	65	56	16.07
20	70	61	14.75
40	95	86	10.47
60	115	107	7.48
80	230	114	101.75
100	165	107	54.21
120	150	118	27.12
140	140	109	28.44
160	145	107	35.51
180	135	112	20.54
200	125	100	25.00
220	130	81	60.49
240	115	81	41.98
260	195	71	174.65
280	110	82	34.15
300	120	73	64.38
320	115	60	91.67
340	105	61	72.13
350	95	66	43.94

D. OpenAI’s ChatGPT 4.0

ChatGPT is a generative artificial intelligence chatbot developed by OpenAI. Released in November 2022, ChatGPT became one of the most prominent AI tools. The latest models of ChatGPT allow users to interact with it through text, audio, and image prompts. When extracting data from

visuals, ChatGPT performs well enough for trends to be made out. However, when visuals are complex and contain overlapping elements, ChatGPT fails to extract useful information from them. When prompted to extract data from the solid red line in Figure 1, ChatGPT was unable to produce numerical data from the figure.

Fig. 2. ChatGPT’s reproduction of the solid red line.

The reproduction of the solid red line shows that the ChatGPT model confused multiple elements of the visual. One notable error is the mislabeling of the axes, which could lead to misinterpretation of the data. While the overall trends of the solid red line are accurately captured, including its rises and falls, ChatGPT failed to distinguish overlapping visual elements. Specifically, the model misclassified portions of the solid red line that intersected with the dotted red line. ChatGPT combined the two distinct data series wherever the points converged. This difficulty illustrates the challenge ChatGPT and other large language models face in analyzing complex visualizations with overlapping elements.

E. DeepSeek

DeepSeek is another large language model capable of extracting information. DeepSeek is a Chinese AI startup that has employed innovative techniques and optimization strategies compared to other models [9]. DeepSeek uses a multi-stage vision encoder and a Mixture-of-Experts text decoder to extract data. DeepSeek divides the graph into 16x16 patches, outlines different parts of the graph for layout, uses a convolutional compressor to reduce the number of tokens found in the graph, and finally uses a Contrastive Language-Image Pre-training framework with a Vision Transformer architecture to make sense of the information found in the graph [10]. This framework works well with textual documents, but doesn’t work with visuals. This represents the issue many large language models face in extracting data. Most LLMs are explicitly designed for textual documents and do not support graph-based processing or extract useful information.

IV. COMPARING EXTRACTION CAPABILITIES

As shown in Table 1, there are apparent disparities between manually collected data and data generated

by large language models. The percent error between Google Gemini's data extractions ranges from 1.35% to 29.63%.

With Grok, the disparities grow larger. This large language model places greater emphasis on textual analysis, resulting in much higher percent errors. Among all LLMs, Claude performed best. This is primarily due to its advanced Optical Character Recognition software, which enables the model to interpret complex visual layouts and understand the spatial relationships among axes.

While the LLMs did not perform overwhelmingly poorly, there are still many issues with using an LLM to extract data from visualizations. The first challenge is that the LLM cannot extract data at each minute in the graph, whereas this is straightforward to do manually with WebPlotDigitizer. This issue may arise from the challenge that LLMs face in discerning information from complex charts with multiple elements. Google attempted to address this issue, but Gemini 2.5 Pro remains affected [11]. Additionally, it is difficult for LLMs such as Gemini 2.5 Pro to link multiple visual components, including the color, line style, and positioning of data points [12]. Another issue with the LLM's extraction is its overall accuracy. The Gemini's average error percentage is 9.71%, indicating that its results were not sufficiently accurate for use. The average error percentage increased with other models.

V. CONCLUSION

While Large Language Models can perform a vast array of tasks, these jack-of-all-trades systems have not mastered them all. Data extraction from complex graphs is an example of a task for which LLMs cannot consistently achieve high accuracy. Although these models can capture general trends in data visualizations, LLMs often struggle to provide the precise quantitative analysis required for subsequent tasks.

This limitation stems from LLMs' lack of direct access to the underlying numerical data in graphs. Instead, these models must interpret visual information through pattern recognition, which easily leads to approximations and errors. These errors are especially frequent in regions where graphical elements overlap or where there are

variations in color or line style. This challenge was particularly evident in the outputs from Grok and ChatGPT, which produced the worst results due to the complexity of the input.

Given this disparity between LLMs' descriptive capabilities and the precision often required in scientific contexts, traditional methods are best for accurate data extraction. Manual digitization, specialized software tools such as WebPlotDigitizer, or direct assistance from data analysts remain the best methods for understanding data.

As LLMs continue to improve, they will undoubtedly offer increasing value in summarizing large datasets, suggesting possible interpretations, and identifying broader patterns that might otherwise go unnoticed. Despite these improvements, it is essential to recognize the boundaries of Large Language Models' capabilities and to strategically complement their use with more precise analytical techniques when accuracy is critical.

VI. WHY THIS PROJECT?

This project was chosen because of the author's growing interest in Large Language Models, which have quickly emerged as revolutionary tools in the field of artificial intelligence. This fascination with LLMs stems from their incredible ability to process vast amounts of textual data and generate human-like responses. These capabilities enhance efficiency across a wide array of tasks, including data entry, data analysis, and communication.

In this project, the author seeks to examine the shortcomings of LLMs. This report demonstrated that data extraction from complex data visualizations is a task that LLMs can't perform with high accuracy. By highlighting this limitation, the author aims to understand the inner workings of LLMs that cause these fallbacks and help improve these areas.

VII. REFERENCES

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Manually Extracted Data via WebPlotDigitizer

t [min]	G(t) [mg/dl]	t [min]	G(t) [mg/dl]	t [min]	G(t) [mg/dl]	t [min]	G(t) [mg/dl]
1	56	2	56	3	56	4	56
5	56	6	57	7	57	8	57
9	57	10	57	11	58	12	58
13	60	14	60	15	60	16	60
17	60	18	60	19	61	20	61
21	62	22	62	23	62	24	62
25	64	26	64	27	65	28	66
29	66	30	67	31	67	32	69
33	69	34	69	35	70	36	71
36	77	37	78	38	82	39	83
40	86	41	88	42	91	43	92
44	98	45	99	46	103	47	103
48	105	49	107	50	107	51	107
52	107	53	107	54	107	55	107
56	107	57	108	58	108	59	107
60	107	61	107	62	107	63	107
64	107	65	107	66	107	67	107
68	108	69	108	70	109	71	109
72	109	73	111	74	111	75	112
76	112	77	112	78	112	79	113
80	114	81	115	82	115	83	116
84	116	85	116	86	116	87	117
88	117	89	117	90	117	91	117
92	116	93	116	94	115	95	113
96	112	97	111	98	109	99	108
100	107	101	104	102	103	103	100
104	96	105	96	106	94	107	94
108	94	109	96	110	98	111	99
112	101	113	104	114	105	115	107
116	109	117	109	118	112	119	113
120	118	121	118	112	121	123	121
124	121	125	120	126	118	127	118
128	117	129	116	130	116	131	116
132	115	113	115	134	113	135	112
136	112	137	112	138	112	139	111

140	109	141	108	142	108	143	108
144	108	145	107	146	107	147	107
148	105	149	105	150	107	151	107
152	107	153	107	154	107	155	107
156	107	157	107	158	107	159	107
160	107	161	107	162	105	163	105
164	105	165	105	167	105	168	105
169	107	170	107	171	107	172	107
173	107	174	108	175	108	176	109
177	109	178	111	179	111	180	112
181	112	182	111	183	111	184	109
185	108	186	108	187	108	188	107
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193	103	194	101	195	101	196	101
197	101	198	101	199	100	200	100
201	99	202	99	203	98	204	98
205	95	206	95	207	94	208	91
209	91	210	91	211	91	212	91
213	91	214	90	215	87	216	87
217	86	218	84	219	82	220	81
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249	75	250	74	251	73	252	71
253	71	254	71	255	70	256	70
257	70	258	70	259	70	260	71
261	73	262	73	263	74	264	75
265	78	266	79	267	82	268	82
269	82	270	82	271	82	272	83
273	83	274	83	275	83	276	82
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281	82	282	82	283	82	284	81
285	81	286	81	287	79	288	79
289	78	290	78	291	78	292	77
293	77	294	77	295	75	296	75

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301	73	302	71	303	71	304	71
305	70	306	69	307	69	308	67
309	66	310	66	311	65	312	64
313	64	314	64	315	62	316	62
317	62	318	61	319	61	320	60
321	58	322	58	323	57	324	56
325	54	326	53	327	53	328	53
329	53	330	53	331	53	332	53
333	56	334	56	335	57	336	58
337	60	338	60	339	61	340	61
341	62	342	64	343	64	344	64
345	64	346	65	347	65	348	65
349	66	350	66				